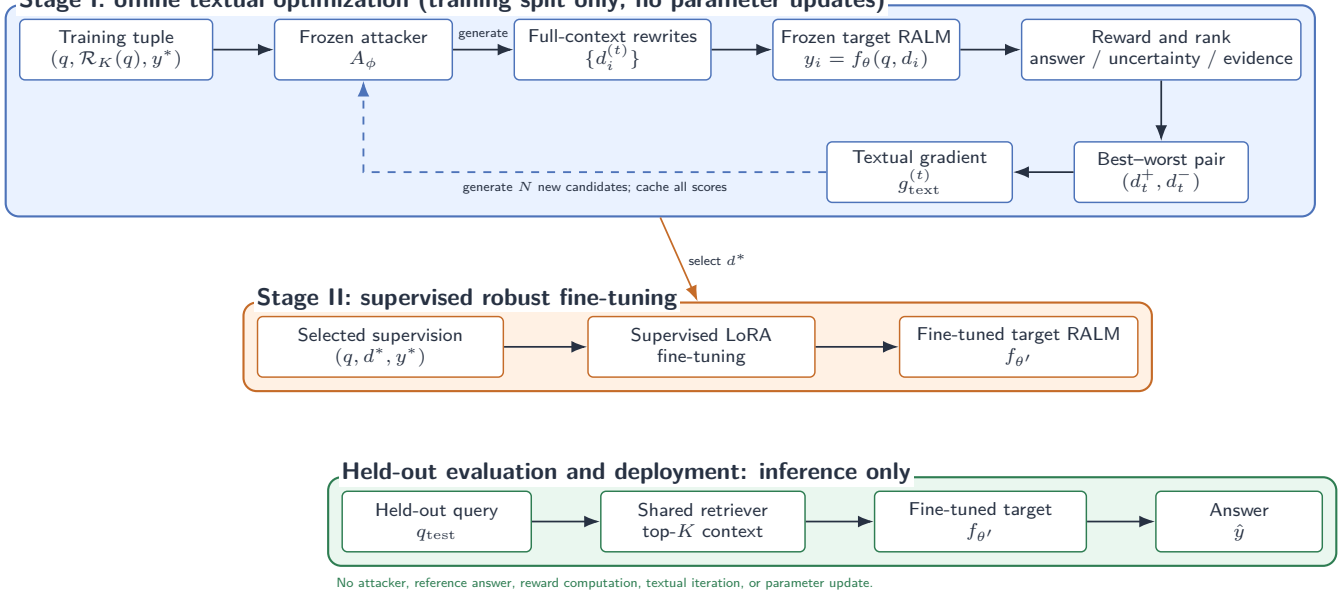
Stage I: offline textual optimization (training split only; no parameter updates)
Training tuple
$(q, \mathcal{R}_K(q), y^*)$
Frozen attacker
A_ϕ
generate
Full-context rewrites
$\{d_i^{(t)}\}$
Frozen target RALM
$y_i = f_\theta(q, d_i)$
Reward and rank
answer / uncertainty / evidence
Best–worst pair
(d_t^+, d_t^-)
Textual gradient
$g_{\text{text}}^{(t)}$
generate N new candidates; cache all scores
select d^*
Stage II: supervised robust fine-tuning
Selected supervision
(q, d^*, y^*)
Supervised LoRA
fine-tuning
Fine-tuned target RALM
$f_{\theta'}$
Held-out evaluation and deployment: inference only
Held-out query
q_{test}
Shared retriever
top-K context
Fine-tuned target
$f_{\theta'}$
Answer
$\hat{y}$
No attacker, reference answer, reward computation, textual iteration, or parameter update.